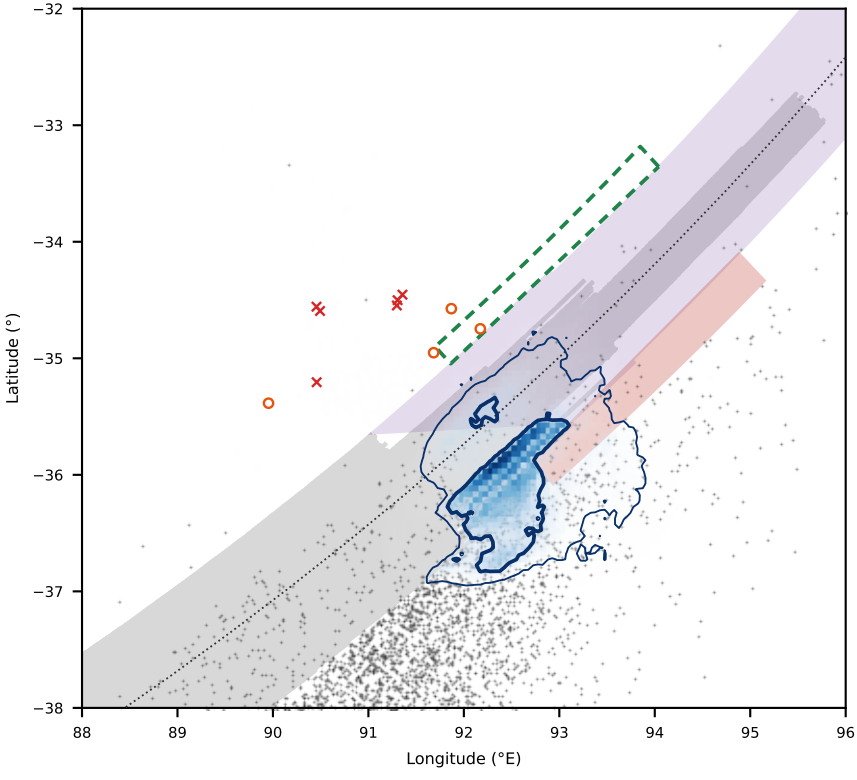
If the imaged objects are from 9M-MRO: where the debris entered the sea — 00:19 R600 BTO Only, no further APU log-on before impact (declared variant)
Pléiades objects + all four COSMO-SkyMed contacts, one debris field
after Phase 2 + Bluefin-21 + OI 2018 + 2025-26 (grade C)




| strata weights | re-weighted (drawn) | fixed |
|---|---|---|
| 50 % region | 7,960 km² | 7,960 km² |
| 90 % region | 32,009 km² | 32,009 km² |
| mean | 35.98 S 92.55 E | 35.98 S 92.55 E |
| outside past searches | 65.0 % | 65.0 % |
| in OI north-west band | 0.1 % | 0.1 % |
| searches leave, under H | 0.309 | 0.309 |
| tension, ln S (p) | -0.75 (p 0.23) | -0.74 (p 0.23) |
| mean shift from flight PDF | 59 NM | 58 NM |

- probability density under H (relative)
- 50 % region under H
- 90 % region under H
- impact PDF without H (sampled points)
- ATSB Phase 2 2014-17 (official)
- Ocean Infinity 2018 (grade C)
- OI 2025-26 south-east band (grade C)
- OI north-west band (not searched; grade C)
- 7th arc, FL400
- Pléiades rating-5 objects
- COSMO-SkyMed radar contacts (all four)

Source: impacts next-run-c (strata next-c-free, next-c-repro-radar, next-c-descent-climb, next-c-routes; provenance read from next-c-free) (8 seeds pooled with equal weight; source run runs/C8-next-c-free, code 097a5d77); prior track 289.7 deg (sd 1.0); config chain (run.json): davey2016-inmarsat + no-exhaustion-prior + s1-factor + s2-temperature + s3-internal + s4-ceiling + s5-hard-reject + s6-tanks + reference-snapshots + radar-full + s7-single-engine + s8-hold-taper + inop-flow-fix + free + server-c-free + hypotheses/end-of-flight/smoke/snapshot-m0011.toml + hypotheses/end-of-flight/smoke/terminal.toml + hypotheses/end-of-flight/full/reference-289.toml + hypotheses/end-of-flight/full/descent-idle-floor.toml + hypotheses/end-of-flight/full/seed-1.toml + hypotheses/end-of-flight/smoke/v2-broad.toml + hypotheses/end-of-flight/full/family-b-ditching.toml + hypotheses/end-of-flight/full/residual-bank-boeing.toml; tempered sampler (m1839, m1941, m2041, m2141, m2241, m0011, 16 stages); particles per mode [1000000, 500000, 2500000, 500000, 2500000]; BFO bias (run.json bfo_bias): {"mean_hz":150.0,"sd_hz":25.0}.

What was run: 00:19 R600 BTO Only: R600 BTO (18,400 µs) scored; no BFO; log-on not from fuel exhaustion (no lag term); aircraft transmitting at 00:19:37, not powered at 01:15:56, and no further APU log-on before impact under end of flight's log-on lag model (end of flight's 'silent' constraint; declared variant, ruling ~19:10 UTC B (c)) [end-of-flight arm: r600-bto+silent]. Strata mixed by P(family) re-weighted by the 00:19 evidence (ruling C; end of flight's family-evidence-run-c.json, key '00:19 R600 BTO Only +silent': next-c-free 0.5983, next-c-repro-radar 0.2371, next-c-descent-climb 0.1410, next-c-routes 0.0236); the core's fixed P(family) (next-c-free 0.5890, next-c-repro-radar 0.2520, next-c-descent-climb 0.1350, next-c-routes 0.0240) is shown beside it. The search then re-weights strata by their own evidence.

Labels: run C: core 66d77b5 option C (free 7M x 8 seeds, others 3.5M x 8), split-half NOT converged (floor 0.924: free 0.918, Davey dynamics + radar 0.910, descent-climb 0.892, routes 0.850); internal-v1 fuel with one-engine flow correction; EoF run C recipe (PROVISIONAL physics; descents above 6,500 ft/min and unloading not reachable); wide Pléiades grid 78-115 E, 45-5 S.

Search: the searched-areas module's own per-impact likelihood (point target, rho = 0.05, shared). Phase 2 (q 0.945), Bluefin-21 (q 0.9) official; OI 2018 (q 0.9, coverage 0.889) and OI 2025-26 south-east band (q 0.9, coverage 0.7808) are community tracings, grade C; the north-west band is drawn only, never negative evidence.

Under H: 12 rating-5 Pléiades objects in 6 clusters (3 km), equal weights; COSMO-SkyMed contacts equally weighted, dawn 20 / dusk 21 Mar passes equal (unverified source, time, footprint); 'one debris field' = likelihoods multiplied per ocean model; GLORYS12 + ERA5 and GlobCurrent daily + ERA5 equal weight; measured transport error (~100-110 km per component at 15 d); windage 0-5 %; Pléiades and COSMO errors independent (rho = 0).

Conditional on H (the objects are debris from 9M-MRO); no Bayes factor or P(H | data).

Background: the impact PDF without H (flight posterior alone, after the same searches), as 6,000 points drawn in proportion to its probability from 0.05 deg grid cells and jittered within each cell. 42 % of it lies inside this frame.